[Total No. of Questions - 9] [Total No. of Printed Pages - 3]

May-26-0478

CSPE-611(vi) (Advanced Machine Learning)

B.Tech. 6th (NEP)

Time : 3 Hours Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all by selecting one question each from section A, B, C and D. Section-E is compulsory.

SECTION - A

1. (a) Define an Artificial Neural Network. Using suitable example(s), discuss the biological inspiration behind ANN and how the architecture of a synthetic neuron mimics a biological one. (6)

(b) Describe the structural differences between a Perceptron and an MLP. Why is a hidden layer necessary for solving non-linear problems like XOR? Explain using suitable example and/or diagram. (6)

2. (a) Describe the complete Backpropagation algorithm. Explain how does the "Error Signal" propagate from the output layer back to the input layer, using a suitable example. (6)

(b) Compare and contrast Sigmoid, Tanh, and ReLU activation functions. What are the pros and cons of each in deep networks? (6)

SECTION - B

3. (a) Explain the difference between Frequentist and Bayesian approaches to probability. How does Bayesian learning incorporate "Prior" knowledge? Explain using a suitable example. (6)

(b) Explain the Naïve Bayes algorithm for classification. What is the "conditional independence assumption," and why is it used? Using an example, explain the issue that may result due to the conditional independence assumption. (6)

4. (a) Explain the principle of MLE. How does it differ from Maximum A Posteriori (MAP) estimation? Explain using an example. (6)

(b) Describe the Brute-Force Bayesian concept learning algorithm. Why is it computationally expensive? Describe how Naïve Bayes is applied to document classification. (6)

SECTION - C

5. (a) Explain how a concept is represented as a tree. What do the internal nodes, branches, and leaf nodes signify? Explain using a suitable example. (6)

(b) Define Entropy in the context of information theory. Provide the formula and explain what a high vs low entropy value indicates about a dataset. How is Information Gain used for attribute selection? (6)

6. (a) Discuss why Decision Trees are highly susceptible to overfitting. Using an example, describe the different signs of an overfitted tree. (6)

(b) How do decision tree algorithms handle noise in the training data and missing attribute values? Analyze the time complexity for building a decision tree and the time complexity for classifying a new instance. (6)

[P.T.O.]

SECTION - D

7. (a) Define Reinforcement Learning. Explain the Agent-Environment interface and the concept of "Exploration vs Exploitation" using an example scenario. Why is function approximation (like using a Neural Network) necessary for RL in high-dimensional state spaces? (6)

(b) Explain the core philosophy of Ensemble Learning. What are learners in this type of learning? Why does combining multiple "Weak Learners" result in a "Strong Leaner"? (6)

8. (a) Explain the concept of Boosting. Describe the sequential learning process where each new model corrects the errors of its predecessor. What type of weak learners are present in boosting? (6)

(b) Using a suitable example, explain Random Forests in detail. Show how it combines Bagging with "Feature Randomness". (6)

SECTION - E (Compulsory)

9. (a) What is the significance of the "Bias" term in a neural network layer?

(b) What is the Symmetry Breaking problem?

(c) What is the MAP Hypothesis?

(d) What is a latent variable in the context of Bayesian learning?.

(e) Define Stop Criteria in the context of tree induction.

(f) Define the Policy (π) in Reinforcement Learning.

(6×2=12)